

FC7xxx TRGSEL Integration Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/07/14	Initial release for MCAL V0.1.0
0.3	2023/10/20	Updated for MCAL V0.3.0
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

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Chapter 1 Introduction

1.1 Introduction

This integration manual describes the integration requirements for the TRGSEL module.

Chapter 2 Building

2.1 Dependencies on Other Modules

Module configuration dependency

- Trgsel: This module provides the configurations of TRGSEL for TRGSEL input and output.
- Common: This module is the basic module which used to choose the chip.

Module build dependency

- Common: This module provides common type for other modules.
- Det: This module is necessary for enabling Development error detection.

2.2 Files Required for Compile

Trgsel module files:

- MCAL/Src/Trgsel/Src/CDD_Trsgel.c
- MCAL/Src/Trgsel/Src/Trgsel_Drv.c
- MCAL/Src/Trgsel/include/CDD_Trsgel.h
- MCAL/Src/Trgsel/include/TrgSel_Drv.h
- MCAL/Src/Trgsel /include/TrgSel_MemMap.h
- MCAL/Src/Trgsel/include/TrgSel_Reg.h
- MCAL/Src/Trgsel/include/TrgSel_Types.h
- MCAL/Src/Trgsel/include/TrgSel_Version.h

- _MCAL_XXXX_generate_XX/src/CDD_Trsgel_PBcfg.c
- _MCAL_XXXX_generate_XX/src/CDD_Trsgel_Cfg.c
- _MCAL_XXXX_generate_XX /include/CDD_Trsgel_Cfg.h

Common module files:

- MCAL/Src/Common/include/Mcal.h
- MCAL/Src/Common/include/Platform_Types.h
- MCAL/Src/Common/include/Compiler_Cfg.h
- MCAL/Src/Common/include/CompilerDefinition.h
- Compile Related Files/Common/Include/Compiler.h
- Compile Related Files/Common/Include/Std_Types.h

Det module files:

- Det.h

2.3 Add Plug-ins

TRGSEL module plug-ins are developed for EB tresos Studio, so, to use TRGSEL plug-ins on the EB tresos Studio, the user needs to add the TRGSEL module to the EB plug-ins folder first.

- 1) Copy the TRGSEL module (MCAL/EB_Plugins/eclipse/plugins/ Trgsel) folder to EB tresos plug-ins(EB/tresos/plugins/) folder.
- 2) Set the TRGSEL module output location folder for the generated source file and header files.
- 3) Use EB tresos to configure the TRGSEL module and generate source and header files.

Chapter 3 Memory

3.1 Sections in Memory Map

Section name	Section type	Description
TRGSEL_START_SEC_CONFIG_DATA_8 TRGSEL_STOP_SEC_CONFIG_DATA_8 TRGSEL_START_SEC_CONFIG_DATA_16 TRGSEL_STOP_SEC_CONFIG_DATA_16 TRGSEL_START_SEC_CONFIG_DATA_32 TRGSEL_STOP_SEC_CONFIG_DATA_32	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (data).
TRGSEL_START_SEC_CONFIG_DATA_UNSPECIFIED TRGSEL_STOP_SEC_CONFIG_DATA_UNSPECIFIED	Configuration Data	Start and stop of Memory Section for Config Data.
TRGSEL_START_SEC_CONST_BOOLEAN TRGSEL_STOP_SEC_CONST_BOOLEAN TRGSEL_START_SEC_CONST_8 TRGSEL_STOP_SEC_CONST_8 TRGSEL_START_SEC_CONST_16 TRGSEL_STOP_SEC_CONST_16 TRGSEL_START_SEC_CONST_32 TRGSEL_STOP_SEC_CONST_32 TRGSEL_START_SEC_CONST_UNSPECIFIED TRGSEL_STOP_SEC_CONST_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read-only (rodata).
TRGSEL_START_SEC_CODE TRGSEL_STOP_SEC_CODE TRGSEL_START_SEC_RAMCODE TRGSEL_STOP_SEC_RAMCODE TRGSEL_START_SEC_CODE_AC TRGSEL_STOP_SEC_CODE_AC	Code	Start and stop of memory Section for Code(text).
TRGSEL_START_SEC_VAR_NO_INIT_BOOLEAN TRGSEL_STOP_SEC_VAR_NO_INIT_BOOLEAN TRGSEL_START_SEC_VAR_NO_INIT_8 TRGSEL_STOP_SEC_VAR_NO_INIT_8 TRGSEL_START_SEC_VAR_NO_INIT_16 TRGSEL_STOP_SEC_VAR_NO_INIT_16 TRGSEL_START_SEC_VAR_NO_INIT_32 TRGSEL_STOP_SEC_VAR_NO_INIT_32 TRGSEL_START_SEC_VAR_NO_INIT_UNSPECIFIED TRGSEL_STOP_SEC_VAR_NO_INIT_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are never cleared and never initialized by startup code (bss).
TRGSEL_START_SEC_VAR_INIT_BOOLEAN TRGSEL_STOP_SEC_VAR_INIT_BOOLEAN TRGSEL_START_SEC_VAR_INIT_8 TRGSEL_STOP_SEC_VAR_INIT_8	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These

Section name	Section type	Description
TRGSEL_START_SEC_VAR_INIT_16 TRGSEL_STOP_SEC_VAR_INIT_16 TRGSEL_START_SEC_VAR_INIT_32 TRGSEL_STOP_SEC_VAR_INIT_32 TRGSEL_START_SEC_VAR_INIT_UNSPECIFIED TRGSEL_STOP_SEC_VAR_INIT_UNSPECIFIED		variables are initialized by startup code (data).
TRGSEL_START_SEC_VAR_NO_INIT_BOOLEAN_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_NO_INIT_BOOLEAN_NO_CACHEABLE TRGSEL_START_SEC_VAR_NO_INIT_8_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_NO_INIT_8_NO_CACHEABLE TRGSEL_START_SEC_VAR_NO_INIT_16_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_NO_INIT_16_NO_CACHEABLE TRGSEL_START_SEC_VAR_NO_INIT_32_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_NO_INIT_32_NO_CACHEABLE TRGSEL_START_SEC_VAR_NO_INIT_UNSPECIFIED_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_NO_INIT_UNSPECIFIED_NO_CACHEABLE		These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are not cacheable and never cleared and never initialized by startup code (bss).
TRGSEL_START_SEC_VAR_INIT_BOOLEAN_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_INIT_BOOLEAN_NO_CACHEABLE TRGSEL_START_SEC_VAR_INIT_8_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_INIT_8_NO_CACHEABLE TRGSEL_START_SEC_VAR_INIT_16_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_INIT_16_NO_CACHEABLE TRGSEL_START_SEC_VAR_INIT_32_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_INIT_32_NO_CACHEABLE TRGSEL_START_SEC_VAR_INIT_UNSPECIFIED_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_INIT_UNSPECIFIED_NO_CACHEABLE		These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are not cacheable and initialized by startup code (data).
TRGSEL_START_SEC_VAR_NO_INIT_SHAREABLE_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_NO_INIT_SHAREABLE_NO_CACHEABLE		These are all the sections used for variables which are shareable. These variables are not cacheable and never cleared and never initialized by startup code (bss).
TRGSEL_START_SEC_VAR_INIT_SHAREABLE_NO_CACHEABLE TRGSEL_STOP_SEC_VAR_INIT_SHAREABLE_NO_CACHEABLE		These are all the sections which are shareable. These variables are not cacheable and initialized by startup code (data).

Chapter 4 Exclusive Area

None. TRGSEL has no global variables to protect.

Chapter 5 Interrupt Service Routine (ISR)

None.

Chapter 6 Error Report

6.1 Det

Function Name	Error Type
TrgSel_ValidateGlobalCall	TRGSEL_E_ALREADY_INITIALIZED_U8 TRGSEL_E_UNINIT_U8
TrgSel_ValidatePtrInit	TRGSEL_E_INIT_FAILED_U8
TrgSel_ValidateVersionInfoCall	TRGSEL_E_PARAM_VINFO_U8
TrgSel_ValidateTrgMuxConfigCall	TRGSEL_E_TRGSEL_LOCKED_U8 TRGSEL_E_INVALID_REGINDEX_U8 TRGSEL_E_INVALID_TRGSEL_SEL_U8
TrgSel_ValidateTrgMuxConfigLockCall	TRGSEL_E_INVALID_REGINDEX_U8

6.2 Dem

None

Chapter 7 Function Calls

7.1 Function Calls during Startup

None

7.2 Function Calls during Shutdown

None

7.3 Function Calls during Wake-up

None

7.4 Function Calls during Runtime

None

Chapter 8 Other Requirements

8.1 Notification, Callback, Callout

None

8.2 Macros

None

Chapter 9 Integration Steps

- 1) Configure the TRGSEL module and generate configuration files. (please refer to Building chapter for details)
- 2) Configure appropriate memory sections in linker file or other. (please refer to Memory chapter for details).
- 3) Build the TRGSEL module with dependent modules.